

6.6.1 Waves in air, fluids and solids

6.6.1.1 Transverse and longitudinal waves

Content	Key opportunities for skills development
Waves may be either transverse or longitudinal. The ripples on a water surface are an example of a transverse wave. Longitudinal waves show areas of compression and rarefaction. Sound waves travelling through air are longitudinal.	
Students should be able to describe the difference between longitudinal and transverse waves.	WS 1.2
Students should be able to describe evidence that, for both ripples on a water surface and sound waves in air, it is the wave and not the water or air itself that travels.	WS 1.2, 2.2

6.6.1.2 Properties of waves

Content	Key opportunities for skills development
Students should be able to describe wave motion in terms of their amplitude, wavelength, frequency and period. The amplitude of a wave is the maximum displacement of a point on a wave away from its undisturbed position. The wavelength of a wave is the distance from a point on one wave to the equivalent point on the adjacent wave. The frequency of a wave is the number of waves passing a point each second.	MS 1c, 3b, c
$period = \frac{1}{frequency}$ $\left[T = \frac{1}{f} \right]$ period, T, in seconds, s frequency, f, in hertz, Hz The wave speed is the speed at which the energy is transferred (or the wave moves) through the medium. All waves obey the wave equation:	MS 1c, 3b, c Students should be able to apply this equation which is given on the <i>Physics equation sheet</i>.
$wave\ speed = frequency \times wavelength$ $[v = f \lambda]$ wave speed, v, in metres per second, m/s frequency, f, in hertz, Hz wavelength, λ, in metres, m Students should be able to: identify amplitude and wavelength from given diagrams	MS 1c, 3b, 3c Students should be able to recall and apply this equation.
describe a method to measure the speed of sound waves in air	AT 1 WS 2.3, 2.4, 2.6, 2.7, 3.1, 3.5
describe a method to measure the speed of ripples on a water surface.	AT 1, AT 4 WS 2.3, 2.4, 2.6, 2.7, 3.1, 3.5

Required practical activity 20: make observations to identify the suitability of apparatus to measure the frequency, wavelength and speed of waves in a ripple tank and waves in a solid and take appropriate measurements.

AT skills covered by this practical activity: physics AT 4.

This practical activity also provides opportunities to develop WS and MS. Details of all skills are given in [Key opportunities for skills development](#) (page 191).